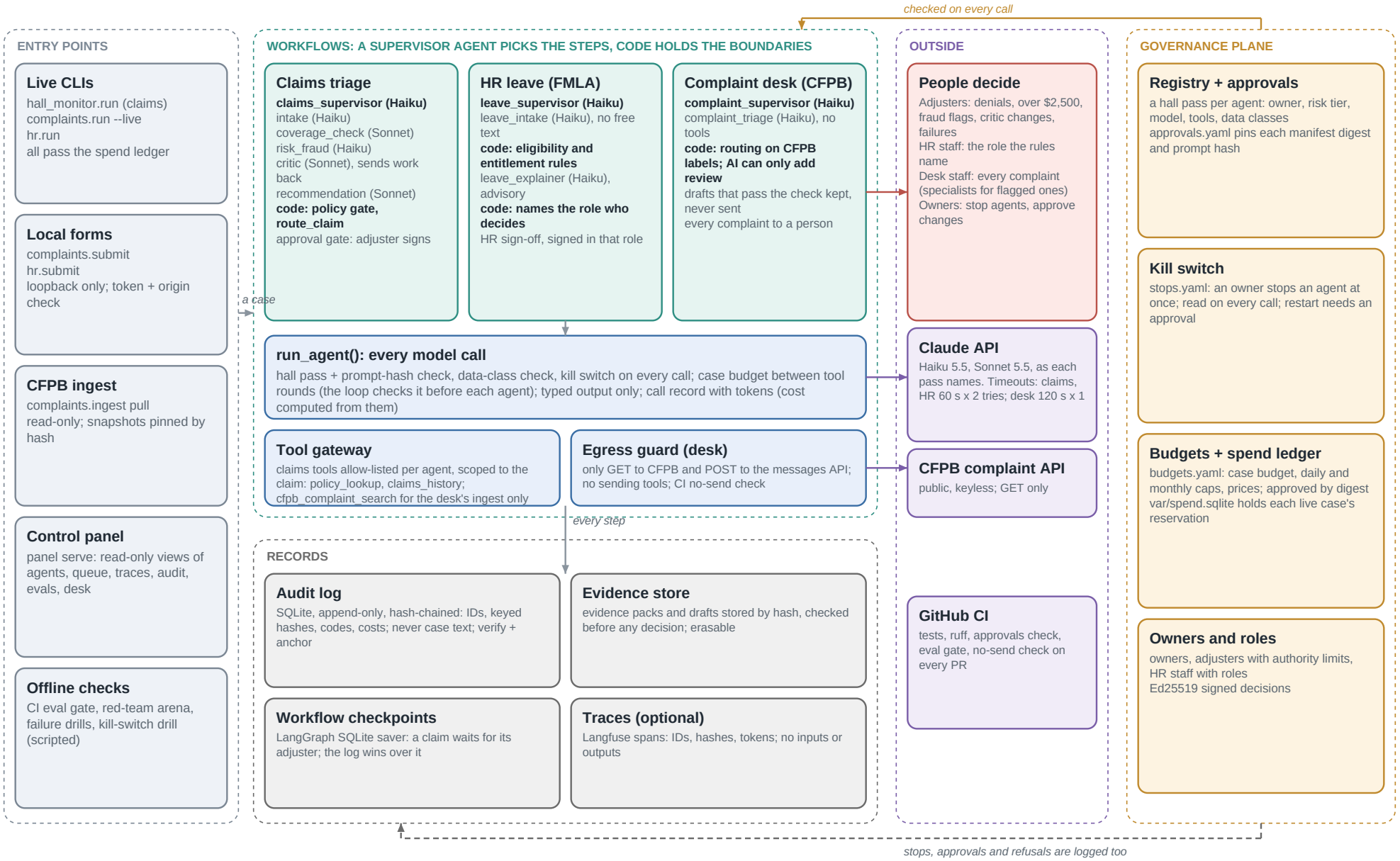
System architecture

Governance sits in the call path, not inside the model. Every model call goes through one function that checks the agent's approved hall pass and the kill switch, and the workflow's loop checks the case budget before every request; tools go through a gateway; routing is plain code; every step lands in a hash-chained audit log.

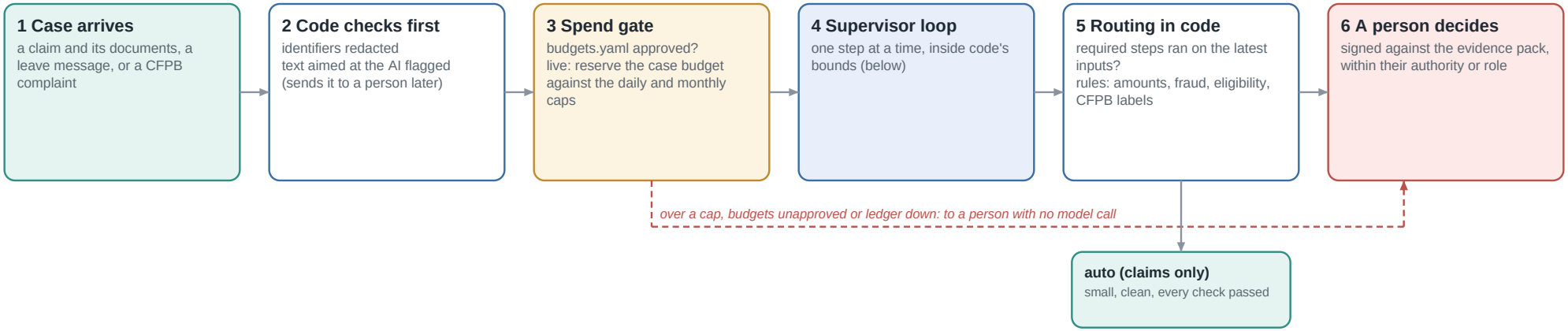
entry points · workflows and agents · shared runtime core · governance plane · records · outside the system · people

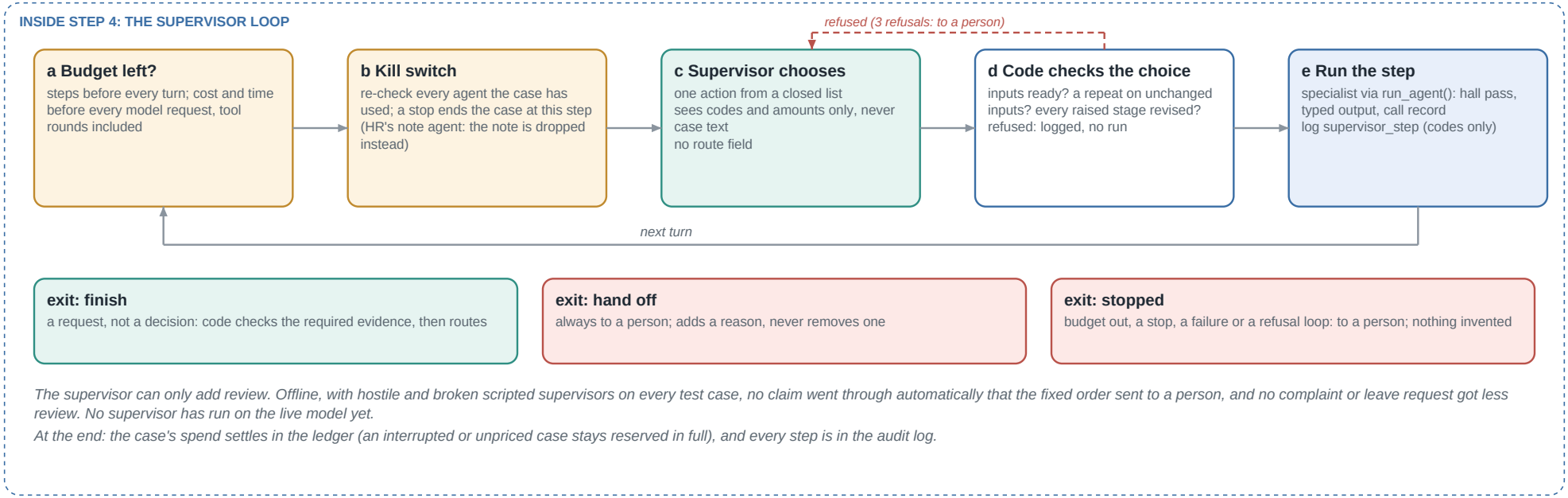


Reference implementation, not a production system. Claims, policies and people are fictional; the complaint desk reads 500 real public CFPB complaints, minimised. As of 13 Nov 2026, from github.com/keeplearnin/hall-monitor.

How one case moves (the same path in all three workflows)

Code checks the case before any model sees it, reserves its budget, then lets the supervisor choose one step at a time. Code checks every choice, and code alone routes the case. Anything that fails, stops or runs out of budget goes to a person; nothing invents a value to keep going.





The supervisor can only add review. Offline, with hostile and broken scripted supervisors on every test case, no claim went through automatically that the fixed order sent to a person, and no complaint or leave request got less review. No supervisor has run on the live model yet.

At the end: the case's spend settles in the ledger (an interrupted or unpriced case stays reserved in full), and every step is in the audit log.

Functionality: three workflows on one set of controls

Each row is the same question asked of each workflow. Agents prepare; code sets the floor; people decide. Status is what has been shown, not what is planned.

	Claims triage	HR leave (FMLA)	Complaint desk (CFPB)
Input	Travel-insurance claim: narrative, receipts, doctors' letters (synthetic)	Employee's leave message (synthetic), employee and company records	Real public CFPB complaints (500 pinned, minimised), or a typed-in one from the form
Agents (model)	supervisor, intake, risk_fraud (Haiku); coverage_check, critic, recommendation (Sonnet)	supervisor, leave_intake, leave_explainer (all Haiku)	supervisor, complaint_triage (both Haiku)
What the AI does	extracts, checks the policy, spots fraud signs, double-checks, recommends	extracts dates, reason and yes/no facts (no free-text field); writes an advisory note	summary, category, urgency, concerns, a draft acknowledgement
What code decides	policy exists; amounts in the documents; auto only if small and clean	FMLA eligibility and entitlement; the role who must decide	specialist review at high urgency for fraud, identity theft, foreclosure, discrimination, Older American, Servicemember
Who decides	adjuster within their authority: denials, over $2,500, fraud flags, critic changes, injection, failures	HR staff at the role the rules name; denying needs a manager or above	a person works every complaint; nothing resolved automatically
Hard rules	lookups only for this claim's policy; decisions signed against the evidence	the rules run on the first intake; no shopping for an answer	built never to reply to or contact anyone: one allow-listed outbound path, no sending tools; drafts that pass the check kept, never sent
Approved budget	16 steps, $0.25, 300 s a case; $5 a day, $50 a month	8 steps, $0.02, 120 s a case; $1 a day, $10 a month	6 steps, $0.02, 120 s a case; $1 a day, $10 a month
Status	built; live scores from the older fixed order; supervisor scripted only	partial: rules unreviewed by HR or a lawyer; agents not run live	partial: scripted runs only; form proposed; 8 of 18 red-team attacks got harm through

Governance controls at a glance (34 in the register)

Read from controls/controls.yaml, where each control lists its code, tests, evidence, NIST AI RMF and FINOS AIGF mapping, and its gaps. 27 built, 3 built with live results pending, 4 partial. Partial means built with stated limits, usually no live run yet.

WHO MAY RUN, AND ON WHAT

built	HM-01	Agent registry and approvals
built	HM-02	Pinned models, no silent fallback
built	HM-03	Tool gateway with allow-list and claim sco...
built	HM-10	Identifier redaction and data minimisation
built	HM-11	Telemetry without claim content
built	HM-19	Kill switch
built	HM-29	Delegated passes only narrow

DECISIONS STAY WITH PEOPLE

built	HM-04	Routing to a person, decided in code
built	HM-05	Adjuster approval gate
built	HM-07	Critic review with revision history
built	HM-08	Failures stop the claim and go to a person
partial	HM-21	Leave eligibility and routing in code (HR ex...
built	HM-23	Signed HR sign-off within the role the rules...
built	HM-27	Complaint triage, routed in code

AGENTS CHOOSE STEPS, CODE HOLDS BOUNDS

partial	HM-32	The agent picks the steps, code holds the...
built	HM-34	Approved budgets and spending caps
built	HM-16	Cost per claim and economy model routing

RECORDS AND EVIDENCE

built	HM-06	Tamper-evident audit log
built	HM-13	Grounding checks and evidence for every...
built	HM-20	Read-only control panel

TESTING, ATTACKS, FAILURES, FAIRNESS

pending	HM-09	Prompt-injection defences
pending	HM-14	Counterfactual fairness test
pending	HM-15	Adversarial suite
partial	HM-30	Red-team arena
built	HM-31	Failure drills
built	HM-12	Eval harness and published results
built	HM-17	CI checks and the eval gate
built	HM-18	Synthetic data and documented limits

WORKFLOW-SPECIFIC

partial	HM-22	Leave agents inside the shared controls (H...
built	HM-24	Leave submission form outside the read-on...
built	HM-25	No reply, no send
built	HM-26	Complaint ingestion from CFPB
built	HM-28	Complaint desk evals, fairness and panel...
built	HM-33	Complaint submission form outside the rea...